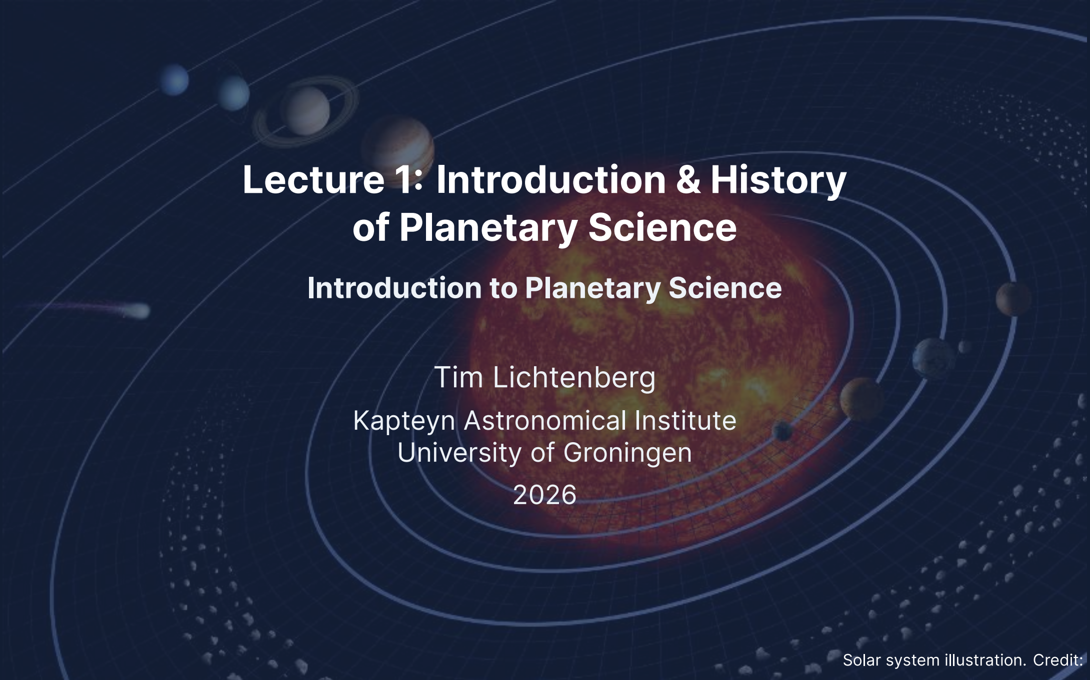
Lecture 1: Introduction & History of Planetary Science

Introduction to Planetary Science

Tim Lichtenberg

Kapteyn Astronomical Institute
University of Groningen

2026



Solar system illustration. Credit: NASA

Learning Objectives

By the end of this lecture, you will be able to:

- Describe the scope of planetary science and its three driving questions
- Explain how the field evolved from antiquity to the space age
- Identify the key properties and classification of solar system bodies
- Derive the stellar mass from planetary orbital parameters

Course Overview

14 lectures + 7 tutorials spanning:

- Formation & orbital dynamics
- Heat, interiors, differentiation
- Atmospheres & surfaces
- Terrestrial planets in detail
- Gas & ice giants, small bodies
- Exoplanets & astrobiology

Assessment: Mid-term (30%) + Final (70%)

Three driving questions

1. How did our solar system form?
2. What makes a planet habitable?
3. Are we alone?

1. A Pale Blue Dot

Earth from Apollo 17 (1972). Credit: NASA/Apollo 17

A Pale Blue Dot

Credit: NASA/JPL-Caltech (1990)

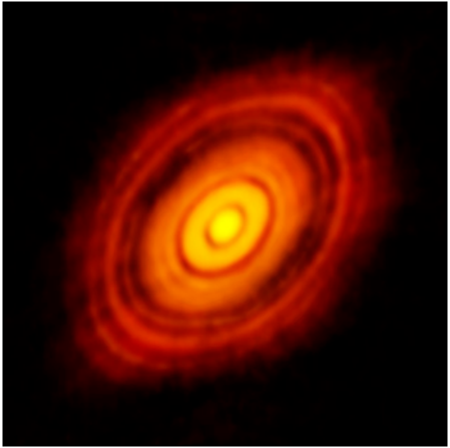
On 14 February 1990, Voyager 1 turned its camera back toward the inner solar system from ~6 billion km (~40 AU).

"Look again at that dot. That's here. That's home. That's us."

Carl Sagan (1994)

Earth appears as a tiny speck, less than a single pixel, suspended in a scattered beam of sunlight.

Question 1: How Did Our Solar System Form?



Credit: ALMA/ESO/NAOJ/NRAO (2014)

- Before 1992: zero confirmed exoplanets
- Today: > 5,700 confirmed exoplanets in > 4,300 systems
- ~0.4–0.6 rocky, habitable-zone planets per Sun-like star

⇒ **Hundreds of millions** of such worlds in the Milky Way

Is our solar system typical, or unusual?

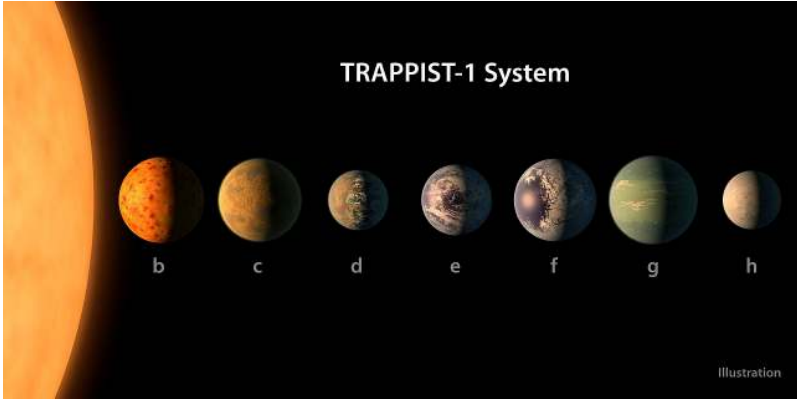
Question 2: What Makes a Planet Habitable?

Credit: NASA/JPL-Caltech (2022)

- Same protoplanetary disk, similar compositions
- All three may have had early liquid water
- Today:
  - **Venus:** 460 °C, 90 bar $\mathrm{CO_2}$
  - **Earth:** Oceans, biosphere
  - **Mars:** Cold desert, <1% atmosphere

Why did they diverge?

Question 3: Are We Alone?



Credit: NASA/JPL-Caltech (2017)

- TRAPPIST-1: seven Earth-sized planets, three in the habitable zone
- JWST is now characterising their atmospheres
- Within your careers, this question may become answerable

2. What Is a Planet?

Pluto in true colour. Credit: NASA/JHUAPL/SwRI (2015)

What Is a Planet?

- Greek: *planetes* (*πλανήτης*) = “wanderer”
- Five visible wanderers: Mercury, Venus, Mars, Jupiter, Saturn
- With the Sun and Moon → seven days of the week
- Planet count unchanged for over two millennia

Telescopic discoveries:

- 1781: Herschel discovers **Uranus**
- 1846: Galle observes **Neptune** (predicted by Le Verrier)
- 1930: Tombaugh discovers **Pluto**

The Pluto Problem

Credit: ESO/L. Calçada/N. Risinger

- Pluto was always an oddity: small, icy, eccentric & inclined orbit
- 1990s–2000s: many large Kuiper Belt objects discovered
- 2005: discovery of **Eris**, comparable in size to Pluto

Either these new objects are also planets, or Pluto is not.

IAU Definition (2006)

Resolution B5. A **planet** in the solar system must:

1. Be in orbit around the Sun
2. Have sufficient mass for self-gravity to achieve hydrostatic equilibrium (nearly round)
3. Have **cleared the neighbourhood** around its orbit

Result:

- 8 planets
- 5 recognised dwarf planets: Ceres, Pluto, Eris, Makemake, Haumea

Criticism:

- "Clearing" is not precisely defined
- Depends on heliocentric distance
- Earth wouldn't clear its zone at Neptune's orbit

Alternative: Geophysical Definition

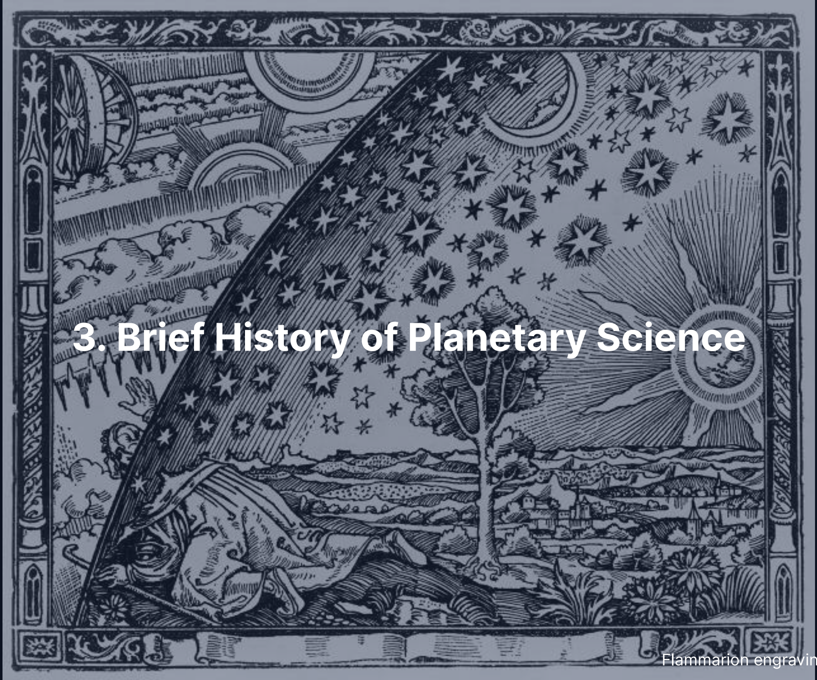
Runyon et al. (2017)

Any body massive enough to achieve **hydrostatic equilibrium** is a planet, regardless of orbital dynamics.

- Under this definition: >100 planets in the solar system
- Includes large moons (Titan, Europa, Ganymede, ...)

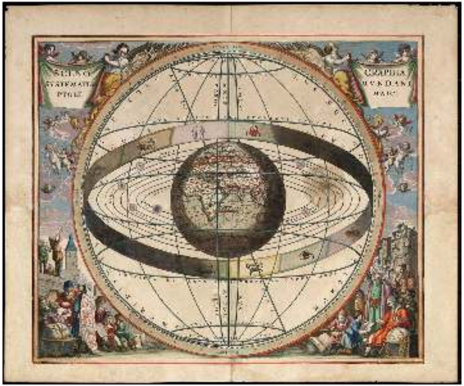
For this course, classification matters less than physics. The solar system contains a continuous spectrum of objects, from dust grains to gas giants, governed by the same physical laws.

3. Brief History of Planetary Science



Flammarion engraving (1888). Public domain

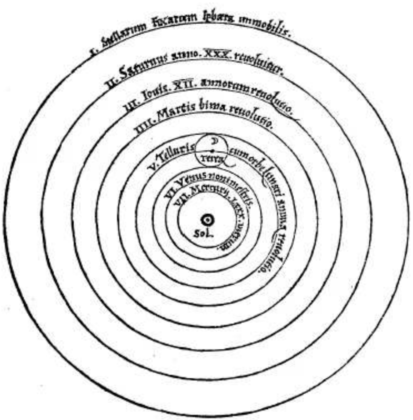
Ancient & Pre-Telescopic Astronomy



Credit: A. Cellarius (1660)

- ~**1800 BCE**: Babylonians track planetary positions
- ~**150 CE**: Ptolemy's epicyclic system, standard for >1000 yr
- **1543**: Copernicus, Sun at the centre
- **1609–19**: Kepler's three laws, circles → ellipses
- **1687**: Newton's *Principia*, universal gravitation explains all three

The Copernican Revolution



Credit: Copernicus, *De revolutionibus* (1543)

- Heliocentric model: Sun (*Sol*) at the centre
- Earth (*Terra*) orbits like any other planet
- Conceptual shift from 1400 years of geocentric cosmology
- Kepler refined circular orbits → **ellipses**
- Newton showed: single law of gravity explains all three of Kepler's laws

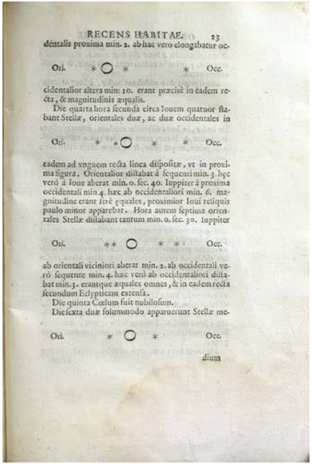
Galileo's Telescope (1610)

Key observations that transformed planetary science:

- Four moons orbiting Jupiter (Galilean moons)
  → Not everything orbits Earth
- Phases of Venus
  → Venus orbits the Sun
- Saturn's rings (could not interpret their structure)
- Mountains and craters on the Moon
  → Celestial bodies are *physical worlds*

Planetary science shifts from pure mathematics to a **physical science**.

Sidereus Nuncius (1610)


RECENS HABITAE. 23

dentalis proxima min. 2. ab hac vero elongabatur oc-

Ori. Occ.

cidentalior altera min: 10. erant præcisè in eadem recta, & magnitudinis æqualis.

Die quarta hora ſecunda circa Iouem quatuor ſtabant Stellæ, orientales duæ, ac duæ occidentales in

Ori. Occ.

eadem ad vnguem recta linea diſpoſitæ, vt in proxima figura. Orientalior diſtabat à ſequenti min. 3. hęc verò à Ioue aberat min. 0. ſec. 40. Iuppiter à proxima occidentali min 4. hæc ab occidentaliori min. 6. magnitudine erant ferè ęquales, proximior Ioui reliquis paulo minor apparebat. Hora autem ſeptima orientales Stellæ diſtabant tantum min. 0. ſec. 30. Iuppiter

Ori. Occ.

ab orientali viciniori aberat min. 2. ab occidentali verò ſequente min. 4. hæc verò ab occidentaliori diſtabat min. 3. erantque æquales omnes, & in eadem recta ſecundum Eclypticam extenſa.

Die quinta Cœlum fuit nubiloſum.

Die ſexta duæ ſolummodo apparuerunt Stellæ me-

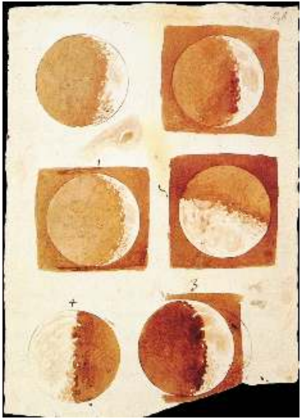
Ori. Occ.

dium

Credit: Galileo Galilei, public domain

- *Sidereus Nuncius* = "Starry Messenger"
- Sketches of Jupiter and its four largest moons (*Medicea Sidera*)
- Recorded changing positions over successive nights
- Direct evidence: not all celestial bodies orbit Earth
- One of the most important publications in the history of science

Galileo's Observations of the Moon



Credit: Galileo (1610)

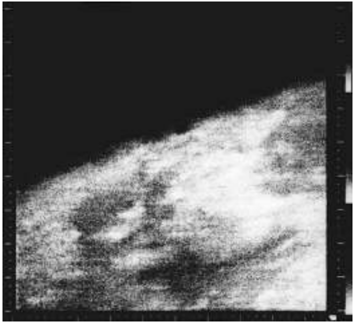
- First telescopic drawings of the Moon (1609–1610)
- Revealed mountains, craters, and rough terrain
- Contradicted the Aristotelian idea of "perfect" celestial spheres
- The Moon is a **physical world** with topography, like Earth

Celestial bodies are not abstract points of light, but tangible worlds that can be studied physically.

The 19th Century: Spectroscopy & Discovery

- Improving telescopes reveal surface features on Mars, Jupiter's Great Red Spot, Saturn's ring structure
- **Spectroscopy** enables remote atmospheric composition measurements for the first time
- Photography enables systematic surveys
- The known solar system steadily expands:
  Uranus (1781) → Neptune (1846) → Pluto (1930)

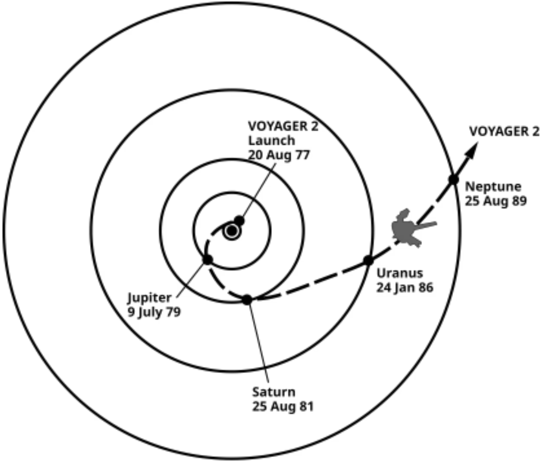
The Space Age Begins



Credit: Mariner 4, NASA/JPL (1965)

- **1962**: Mariner 2 flies past Venus, first planetary flyby
- Revealed Venus's extreme surface temperature (~460 °C)
- **1965**: Mariner 4 returns first Mars close-ups → Cratered, dead world (not canals!)
- **1970**: Venera 7, first surface landing (Venus)
- **1976**: Viking landers, first life-detection experiments on Mars

Voyager Grand Tour



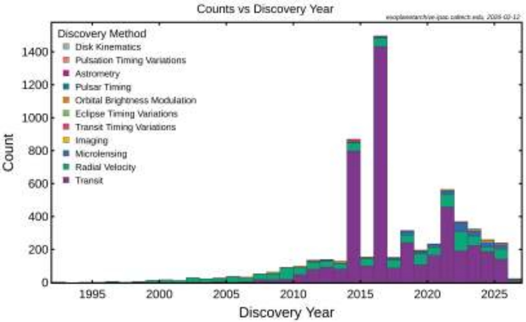
Credit: NASA/JPL, public domain

- Launched 1977
- Exploited a rare planetary alignment (~once every 175 years)
- Gravity assists at each planet
- Jupiter (1979), Saturn (1981), Uranus (1986), Neptune (1989)
- Voyager 1 now in interstellar space

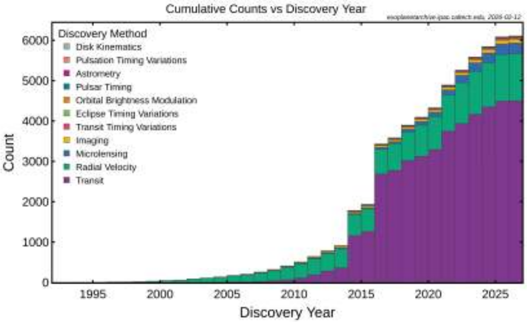
The Exoplanet Revolution

- **1992**: Wolszczan & Frail detect planets orbiting a **pulsar**
  → First confirmed exoplanets
- **1995**: Mayor & Queloz detect **51 Pegasi b**
  Hot Jupiter with 4.2-day orbit → challenged all formation theories
- **2009–2018**: Kepler mission discovers thousands of transiting exoplanets
- **2018–**: TESS surveys the brightest nearby stars
- **2022–**: JWST characterises exoplanet atmospheres

Exoplanet Discoveries Over Time



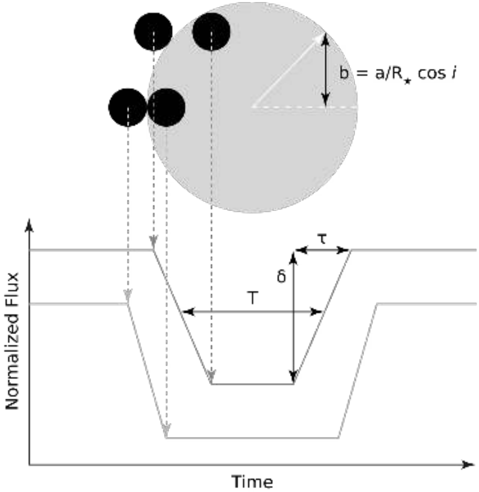
Credit: NASA Exoplanet Archive (2025)



Credit: NASA Exoplanet Archive (2025)

Left: Annual discoveries by detection method. **Right:** Cumulative total. The transit method has dominated since Kepler (2009); we will revisit this in Lecture 13.

The Transit Method



Credit: CielProfond, CC BY-SA 4.0

- Planet passes in front of host star
- Blocks a fraction of starlight:

$$\frac{\Delta F}{F} = \left(\frac{R_p}{R_\star}\right)^2$$

- Gives the planet's **radius**
- Used by Kepler, TESS, JWST
- Has discovered the majority of known exoplanets

Planetary Science Today

Today, planetary science integrates:

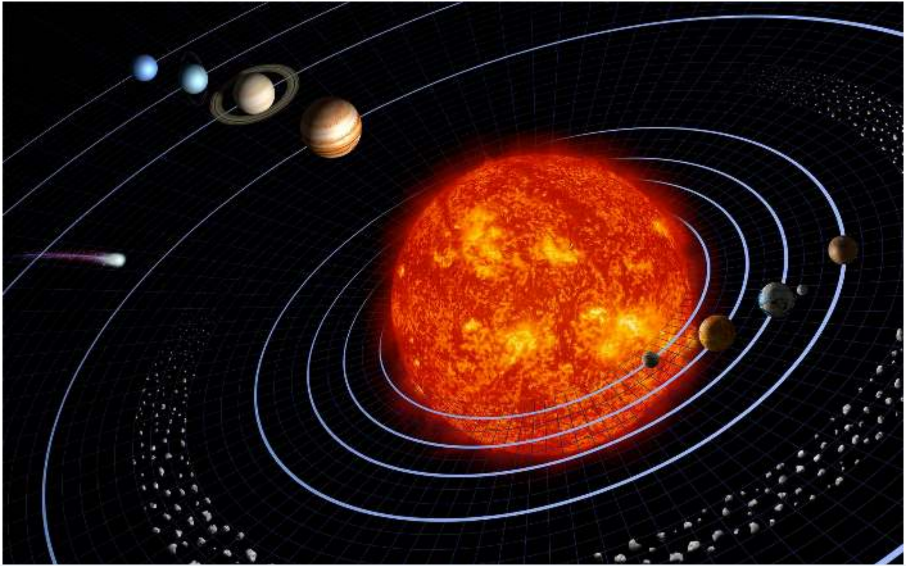
- Astronomy
- Physics
- Chemistry
- Geology
- Atmospheric science
- Biology

It spans scales from **dust grains** in protoplanetary disks to the **demographics of planetary systems** across the Galaxy.

4. Overview of the Solar System

Saturn at equinox, Cassini (2009). Credit: NASA/JPL/SSI

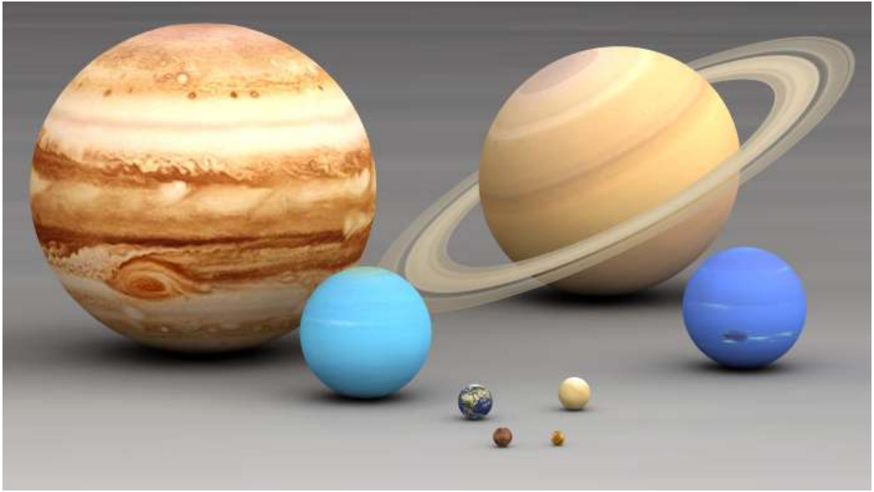
Solar System Architecture



Solar System Zones

- **Inner solar system** (0.4–1.5 AU):
  Mercury, Venus, Earth, Mars: small, dense, rocky
- **Asteroid belt** (2.1–3.3 AU):
  Total mass ~ $4 \times 10^{-4}\ M_{\oplus}$, dominated by Ceres
- **Outer solar system** (5.2–30.1 AU):
  Jupiter, Saturn (gas giants); Uranus, Neptune (ice giants)
- **Kuiper Belt** (30–50 AU):
  Icy bodies: Pluto, Eris, Makemake, ...
- **Oort Cloud** (10^4–10^5 AU):
  Source of long-period comets (not directly observed)

Relative Sizes of the Planets



Top: Jupiter, Saturn, Uranus, Neptune | Bottom: Earth, Venus, Mars, Mercury

Planetary Properties: Inner Solar System

Planet	**Mass** ($M_\oplus$)	**Radius** ($R_\oplus$)	a (AU)	P (yr)	e	ρ ($\mathrm{kg\,m^{-3}}$)
Mercury	0.055	0.383	0.387	0.241	0.206	5427
Venus	0.815	0.949	0.723	0.615	0.007	5243
Earth	1.000	1.000	1.000	1.000	0.017	5514
Mars	0.107	0.532	1.524	1.881	0.093	3934

Credit: NASA Planetary Fact Sheet

All have $\rho > 3900\ \mathrm{kg\,m^{-3}}$: **rock and metal** dominate.

Planetary Properties: Outer Solar System

Planet	**Mass** ($M_\oplus$)	**Radius** ($R_\oplus$)	a (AU)	P (yr)	e	ρ (kg m^{-3})
Jupiter	317.8	11.21	5.203	11.86	0.049	1326
Saturn	95.16	9.45	9.537	29.46	0.054	687
Uranus	14.54	4.01	19.19	84.01	0.047	1271
Neptune	17.15	3.88	30.07	164.8	0.009	1638

Credit: NASA Planetary Fact Sheet

All have ρ < 1700 kg m^{-3}: **gas and ice** dominate.
Saturn is famously less dense than water (ρ_{water} = 1000 kg m^{-3}).

Two Key Patterns

1. **Density decreases with distance**

   Inner: $\rho > 3900\ \mathrm{kg\,m^{-3}}$ (rock & metal)
   Outer: $\rho < 1700\ \mathrm{kg\,m^{-3}}$ (gas & ice)

   ⇒ Reflects the temperature structure of the protoplanetary disk

2. **Mass is concentrated in Jupiter**

   Jupiter > twice the mass of all other planets combined

   ⇒ We will quantify this in the blackboard derivation

Classification of Planets

Terrestrial	Mercury, Venus, Earth, Mars Rocky surfaces, iron cores, thin or no atmospheres (Venus: massive $\mathrm{CO_2}$ atmosphere is the exception)
Gas giants	Jupiter, Saturn Massive $\mathrm{H/He}$ envelopes, no well-defined solid surface Likely rocky/icy cores at high pressure
Ice giants	Uranus, Neptune Interiors dominated by $\mathrm{H_2O}$, $\mathrm{NH_3}$, $\mathrm{CH_4}$ under extreme pressures Topped by $\mathrm{H/He}$ atmospheres

The Terrestrial Planets

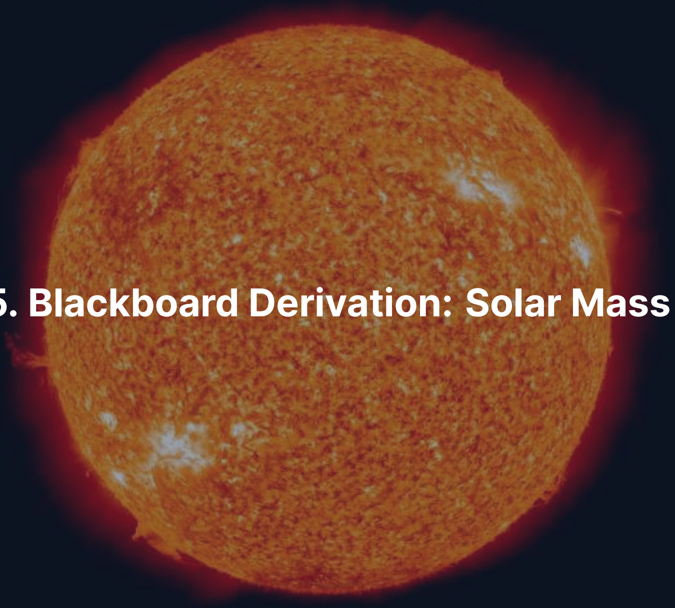
Credit: NASA/JPL, public domain

Mercury, Venus, Earth, Mars at approximate relative scale.
Factor of ~18 in mass, dramatically different evolutionary paths.

Break

Earthrise, Apollo 8 (1968). Credit: NASA

5. Blackboard Derivation: Solar Mass



The Sun, SDO/AIA 171 Å (2010). Credit: NASA/SDO/AIA

Deriving the Solar Mass from Kepler's Third Law

Goal: Estimate $M_\odot$ from Earth's orbital parameters.

Setup: Planet of mass M_p in a circular orbit of radius r around a star of mass M_*.

Gravitational force provides centripetal acceleration:

$$\frac{G M_* M_p}{r^2} = \frac{M_p v^2}{r}$$

where $v = 2\pi r / P$ is the orbital velocity.

Step 1: Substitute Orbital Velocity

Substitute $v = 2\pi r / P$ into the force balance:

$$\frac{G M_* M_p}{r^2} = \frac{M_p}{r} \cdot \frac{4\pi^2 r^2}{P^2}$$

Cancel M_p and simplify:

$$\frac{G M_*}{r^2} = \frac{4\pi^2 r}{P^2}$$

Key insight: The planet's mass cancels; the orbit depends only on the central mass and the orbital radius.

Step 2: Solve for Stellar Mass

Rearranging:

$$\boxed{M_* = \frac{4\pi^2 r^3}{G P^2}}$$

This is **Newton's form of Kepler's third law** (for $M_p \ll M_*$).

Note

For elliptical orbits, replace r with the semi-major axis a.
The general case requires the vis-viva equation (Lecture 2).

Application: Earth's Orbital Parameters

Quantity	Value
Semi-major axis $a_\oplus$	1.496×10^{11} m
Orbital period $P_\oplus$	3.156×10^{7} s
Gravitational constant G	6.674×10^{-11} m^3 kg^{-1} s^{-2}

Result: Solar Mass

$$M_\odot = \frac{4\pi^2 \times (1.496 \times 10^{11})^3}{6.674 \times 10^{-11} \times (3.156 \times 10^7)^2}$$

$$M_\odot \approx 1.99 \times 10^{30}\ \text{kg}$$

Accepted value: $M_\odot = 1.989 \times 10^{30}$ kg

A remarkably accurate estimate from just **two measurable quantities**: a and P.

Planet-to-Star Mass Ratio

The precise form of Kepler's third law gives $M_* + M_p$, not M_* alone.

The approximation $M_* + M_p \approx M_*$ works because:

- Jupiter: $M_\mathrm{J} \approx 318\ M_\oplus \approx 1.90 \times 10^{27}$ kg

$$\frac{M_\mathrm{J}}{M_\odot} \approx 9.5 \times 10^{-4}$$

- Total mass of all 8 planets: $\approx 446\ M_\oplus$

$$\frac{M_\mathrm{planets}}{M_\odot} \approx 1.3 \times 10^{-3}$$

Mass Budget of the Solar System

The Sun contains **99.87%** of the solar system's total mass.

Jupiter alone accounts for **71%** of the planetary mass.

- This extreme concentration of mass in the central star is a **fundamental property** of planetary systems
- Planet formation theory must explain this (Lecture 2)

6. Comparative Planetology

Mars, Rosetta/OSIRIS (2007). Credit: ESA/MPS/UPD/LAM/IAA/RSSD/INTA/UPM/DASP/IDA

Comparative Planetology as a Methodology

- We cannot perform controlled experiments on entire worlds
- **Comparative planetology:** treat planets as a natural set of experiments
- Similar objects subjected to different conditions
- By comparing cases, we isolate which differences arise from:
  - Distance to the Sun
  - Planetary mass
  - Internal activity
  - Historical contingency

Venus, Earth, Mars: Three Divergent Siblings

Property	Venus	Earth	Mars
Surface temperature	460 °C	15 °C	-60 °C
Surface pressure	90 bar	1 bar	0.006 bar
Dominant atmosphere	CO_2 (96.5%)	N_2/O_2 (99%)	CO_2 (95%)
Magnetic field	None	Strong dipole	Remnant crustal
Tectonics	Episodic resurf.	Plate tectonics	Stagnant lid

Formed in the same disk, similar compositions. **Why did they diverge?**

Moons as Natural Experiments

Credit: NASA/JPL/DLR; Io, Europa, Ganymede, Callisto (to scale)

- All orbit Jupiter, differ in tidal heating
- **Io**: most volcanically active body
- **Europa**: subsurface ocean
- **Ganymede**: ocean + magnetic field
- **Callisto**: ancient, undifferentiated
- Same system → vastly different outcomes

7. Observational Techniques

VLT laser guide star, Paranal Observatory. Credit: ESO/Y. Beletsky

Telescopes

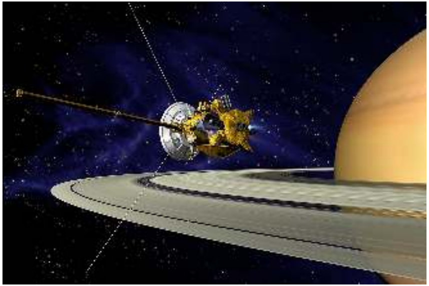
Ground-based:

- Optical imaging: surface and atmospheric features
- Infrared spectroscopy: thermal emission, atmospheric composition
- Radio: subsurface properties, atmospheric dynamics
- Adaptive optics (VLT, Keck): near-space-telescope resolution

Space-based:

- Hubble: atmospheric monitoring of giant planets for decades
- JWST (L2 point): infrared → primary tool for exoplanet atmospheres

Spacecraft Exploration



Credit: NASA/JPL-Caltech

In order of increasing complexity:

Flyby	Voyager 2 at Neptune (1989)
Orbiter	Cassini at Saturn (2004–17)
Lander	Viking 1 on Mars (1976)
Rover	Perseverance on Mars (2021–)
Sample ret.	Hayabusa2 from Ryugu (2020)

Remote Sensing Methods

Spectroscopy Identifies materials by absorption/emission features
→ atmospheric composition, surface mineralogy, thermal properties

Radar Penetrates clouds, maps surface topography
→ Magellan at Venus, MARSIS on Mars (subsurface features)

Gravimetry Maps gravity field from orbital perturbations
→ interior density variations, subsurface mass concentrations

Magnetometry Measures magnetic fields
→ dynamo activity, interior conductivity

Landmark Missions (I)

Mission	Year	Target	Key achievement
Mariner 2	1962	Venus	First planetary flyby
Mariner 4	1965	Mars	First close-up images
Apollo 11	1969	Moon	First crewed landing
Venera 7	1970	Venus	First surface landing
Viking 1 & 2	1976	Mars	First Mars landers
Voyager 1 & 2	1977	Outer	Grand tour
Galileo	1995	Jupiter	Europa’s ocean

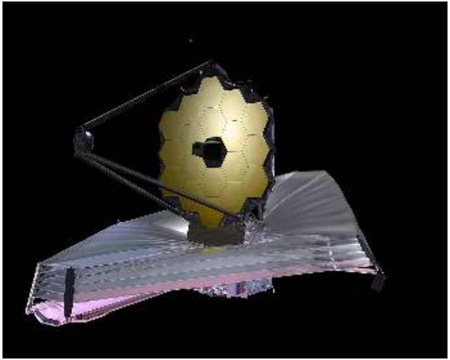
Landmark Missions (II)

Mission	Year	Target	Key achievement
Cassini-Huygens	2004	Saturn	Titan landing, Enceladus plumes
Spirit/Opportunity	2004	Mars	Evidence for past water
New Horizons	2015	Pluto	First Pluto flyby
Hayabusa2	2020	Ryugu	Asteroid sample return
Perseverance	2021	Mars	Sample caching
JWST	2022	Exoplanets	Atmospheric characterisation
Europa Clipper	2024	Europa	Ocean habitability

8. Recent Advances & Outlook

Milky Way, Black Rock Desert. Credit: Steve Jurvetson, CC BY 2.0

James Webb Space Telescope

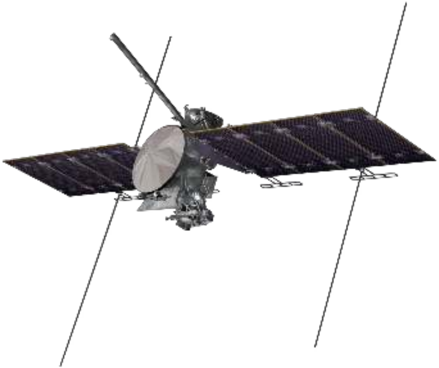


Credit: NASA/GSFC (artist's concept)

- Launched Dec 2021; operating at Sun–Earth L2
- 6.5 m primary mirror, infrared optimised
- First atmospheric characterisation of **rocky exoplanets**
- Thermal emission of TRAPPIST-1 planets
- Do Earth-sized planets around other stars retain atmospheres?

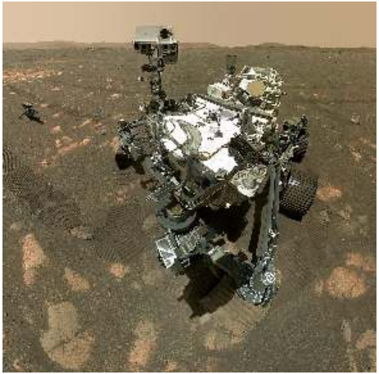
Recent & Upcoming Missions

- **OSIRIS-REx** (2023): Samples from Bennu
  → hydrated minerals, organics
- **Europa Clipper** (2024): En route to Europa
  → subsurface ocean habitability
- **JUICE** (2023→2031): ESA → Jupiter system
- **Dragonfly** (launch ~2028, arrival ~2034): drone on Titan



Credit: NASA/JPL

Surface Exploration: Present & Future



Credit: NASA/JPL-Caltech/MSSS (2021)

Credit: NASA/JHUAPL

Left: Perseverance caching samples in Jezero Crater (Mars). **Right:** Dragonfly will explore Titan's organic-rich surface (launch ~2028, arrival ~2034).

What's Coming: Course Roadmap

Weeks 1–4 (Foundations):

- L2 Formation & orbits
- L3 Heat & energy transport
- L4 Differentiation & magnetospheres
- L5 Atmospheres I
- L6 Atmospheres II
- L7 Surfaces
- *Mid-term exam*

Weeks 5–7 (Applications):

- L8 Interiors
- L9 Earth & Venus
- L10 Mercury & Mars
- L11 Gas & ice giants
- L12 Small bodies
- L13 Exoplanets
- L14 Synthesis & astrobiology

Summary

1. Planetary science asks three driving questions: formation, habitability, life beyond Earth
2. The field evolved from ancient stargazing → telescopes → space missions → exoplanet surveys
3. The solar system has eight planets spanning a huge range in mass, size, and composition
4. From just orbital period and distance, we can measure the Sun's mass: $M_\odot \approx 2 \times 10^{30}$ kg
5. Comparative planetology, which treats worlds as natural experiments, is our most powerful tool
6. We are living in a golden age of planetary exploration

Questions?

Next: **Lecture 2. Planet Formation & Orbital Dynamics**

Lecture notes: `formingworlds.github.io/IntroductionToPlanetaryScience`